

Measure how evolution games imagined safety

Research report: world models, safe reinforcement learning and evolutionary computation

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Summary

The strongest bet for both the thesis and GECCO 2027 is to make world-model exploitation the contribution: evolve compact policies inside the LeWM’s imagination with constraint handling that needs no penalty weight, use exact MuJoCo snapshot branching to measure how quickly selection pressure turns imagined-safe elites into real violations, and show which evolution-native mitigations close that gap at a matched real-evaluation budget. Repeated, differently phrased searches across five research threads found **no paper that evolves a deployable safe policy inside a pixel-trained latent world model and measures imagined against real constraint violations**; the nearest neighbours are GuSS (a 2022/2024 preprint that uses MAP-Elites as a cost-first planner on low-dimensional states and names model uncertainty as its future work), Ha & Schmidhuber’s 2018 dream-versus-real table, and a July 2026 result that larger candidate pools in latent planners pick infeasible plans more often. Much of the surrounding ground is taken: feasibility-first CEM dates from 2018 and 2020 and reappeared at ICLR 2024, so the student’s Safe CEM is a baseline rather than a contribution, conformal robust MPC on a LeWM variant (including Push-T) appeared in June 2026, and Dreamer-style Lagrangian agents and latent Hamilton-Jacobi safety filters are crowded. The lead study should be paired with a cheap real-environment benchmark of evolutionary constraint handling against PPO-Lagrangian on SafetyWalker2dVelocity-v1, which answers the supervisor’s “no hand-crafted penalties” brief directly and exposes a testable bridge: under rank-based selection, ROSARL’s minmax penalty reduces to Deb’s feasibility rule. A safety-indexed MAP-Elites/CMA-MAE archive belongs in December and January as a stretch goal. The binding constraint is time, not compute: the 8 November results freeze is 36 days away, so the thesis core must be the exploitation curve plus one or two mitigations, with the full comparison built for a GECCO deadline that is unpublished but likely falls between 19 and 26 January 2027. Before committing, the student must run the outstanding novelty sweeps, settle the ROSARL citation (the evidence points to RLC 2026, not 2024), and measure local throughput.

Every ingredient is published; the combination is not

The idea of evolving a controller inside a learned model’s dream starts with Ha & Schmidhuber. They trained an **867-parameter linear controller with CMA-ES (population 64, 16 rollouts per candidate)** entirely inside an MDN-RNN dream of VizDoom and then transferred it. Transfer depended on a temperature knob. At $\tau=0.10$ the agent scored **2086±140 in the dream but 193±58 in reality**, below the random policy’s 210. At $\tau=1.15$ it scored 918 in the dream and 1092 in reality, above the 750 solve threshold (Ha & Schmidhuber; NeurIPS 2018). The lineage then split four ways. One branch evolved the world model itself against real-environment fitness (Risi & Stanley, GECCO 2019). A second ran evolutionary or CEM *planning* in a latent model, which is how every JEPA model is used today; LeWM plans with CEM over 300 samples at horizon 5 (EPLS; LeWM). A third used rank-based CMA-ES against uncertain Gaussian-process models in data-efficient robotics (Black-DROPS). The fourth ran quality-diversity in imagination: DA-QD was about **20× more sample-efficient** than model-free QD, but it conceded that real coverage falls below the imagined repertoire because of model error (DA-QD). The research found no 2023–2026 paper that trains a standalone policy by ES, GA or CMA-ES purely inside a pixel-trained RSSM, transformer or JEPA world model and reports real transfer.

Safety narrows the field further. GuSS runs MAP-Elites as a per-step MPC planner inside a learned dynamics model. Its archive replacement rule is lexicographic: lower cost wins, and ties go to higher reward. It was tested on Safe Pendulum, Safe Acrobot and Safety-Gym SafeCar-Goal against CEM, RCEM, Safe-MBPO, CPO and PPO-Lagrangian. Its authors state that “if the model is wrong, it could easily lead the agent to

unsafe states” and list “incorporating model uncertainty with QD” as future work (GuSS v2). GuSS differs from the student’s setting in four ways, and each one is a novelty point: it uses low-dimensional states rather than pixel-trained latents, it plans per step rather than evolving an amortised policy, it has no uncertainty handling, and it has no ground truth for measuring false-safe rates. It also remains an unreviewed preprint.

Older work and the 2024–2026 surge have already claimed several things a naive proposal would otherwise list as contributions. **Feasibility-first elite selection is old.** Wen & Topcu’s constrained cross-entropy method ranks feasible policies by objective and infeasible ones by constraint value (CCE, [NeurIPS 2018](#)). RCE applies the same rule to action sequences using worst-case ensemble cost (RCE), and SafeDreamer’s CCEM uses it inside a world model (SafeDreamer, [ICLR 2024](#)). The student’s Safe CEM therefore has to be presented as a credited baseline. **Conformal safe MPC on LeWM is also taken.** SLS² puts conformal dynamics-error sets, a conformal in-distribution support set and a conformalised constraint classifier into robust MPC on a LeWM variant, using a gradient-based SLS/iLQR optimiser, on Reacher, Cube, Push-T and rope (SLS²). Four more areas are crowded: Dreamer-family Lagrangian agents (PIGDreamer, [ICML 2025](#); Nightmare Dreamer), latent Hamilton-Jacobi safety filters (LatentSafe, [RSS 2025](#); UNISafe, [CoRL 2025](#); LatentCBF), frozen-JEPA failure monitors (FARM), and shields that filter sampled candidates for an action-conditioned JEPA (JEPA shields).

The student’s Push-T diagnosis is still useful, provided it is framed correctly. Attributing 75% of false-safe decisions to imagined dynamics maps directly onto Kim, Nakamura & Bajcsy’s split between **prediction gaps and estimation gaps** (Kim et al.). Citing that split anchors the student’s diagnostic in the latent-safety literature.

The opening is in what 2026’s audits of latent world models have *not* measured. “Proposal overgeneration” shows that raising a latent planner’s candidate pool from 72 to 288 **cut the feasibility of minimum-cost selection from 0.375 to 0.062** on Cube, even though every larger pool contained a feasible sequence (ASAR). ARC-Bench finds that frozen JEPA world models rank candidate actions wrongly and that closed-loop replanning masks the defect (ARC-Bench). The intervention-gap study finds that ensemble disagreement ranks out-of-distribution errors poorly (Intervention gap). Backdoored world models steer CEM and Dreamer-style actors to an attacker’s chosen action (Backdoors). All four results concern goal costs or action ranking. **None measures how a constraint predictor’s false-safe rate scales with selection pressure**, and none tests mitigations that act on the selection operator rather than on the model. That gap belongs naturally to evolutionary computation.

Direction	Closest prior work (status)	Already taken	Open gap found
Evolve policies inside a latent world model	Ha & Schmidhuber 2018 (NeurIPS); DA-QD (ICRA 2022); BFM-QD, QD in a policy latent (NeurIPS 2026 per arXiv)	Unconstrained dream evolution; uncertainty-filtered QD in imagination, without safety	No policy evolution in a pixel-trained or JEPA world model with a measured real transfer, 2023–2026
Safety plus evolution in a learned model	GuSS (preprint); reset-free QD with unsafe-behaviour filtering (GECCO 2023 Companion)	Cost-first archive replacement in a learned model	Uncertainty, latents from pixels, amortised policies, ground-truth false-safe rates
Constrained sampling planners in latent models	CCE, RCE, SafeDreamer CCEM; SLS ² (2026)	Feasibility-first elites; conformal robust MPC on LeWM, including Push-T	Stochastic ranking, ϵ -constraint, NSGA-II and augmented-Lagrangian CMA-ES inside latent MPC; false-safe rate against planner hyperparameters
Optimisers exploiting latent world models	ASAR, ARC-Bench, intervention gap, backdoors (all 2026 preprints)	Goal-cost overgeneration; ranking defects masked by replanning	Constraint-side effect of selection pressure; evolutionary remedies

Direction	Closest prior work (status)	Already taken	Open gap found
EC constraint handling for safe policy search	ECRL (IJCNN 2023); safety-informed mutations (GECCO'22 Companion); AutoCost (AAAI 2023); Safe CMA-ES (GECCO 2024)	Stochastic-ranking ERL on torque constraints; evolved intrinsic costs; safe black-box optimisation on synthetic functions	Pure ES/CMA-ES comparison of Deb rules, stochastic ranking, ϵ -constraint and augmented Lagrangian against PPO-Lagrangian on Safety-Gymnasium; the bridge from ROSARL to feasibility rules
Safety dial or reward-cost front	CCPO (NeurIPS 2023); CDT (ICML 2023); LP3 (CoRL 2021)	A single threshold-conditioned policy; a set of policies from a preference distribution	An evolutionary multi-objective or QD front of reward against violation probability, compared with λ -swept PPO-Lagrangian
Safety-indexed QD archive	Constrained MAP-Elites (Talakat); violations as descriptors (Fioravanzo & Iacca); performance-reproducibility trade-off (Flageat et al.)	Feasible and infeasible populations per cell; violation descriptors on numeric benchmarks	A reward $\times$ cost or violation-probability archive for safe RL; no QD-RL work on Safety-Gym(nasium) beyond GuSS
QD stress-testing of world-model safety	BasinLens (explicitly non-EC); FARM (supervised readout); MADRID (AAMAS 2024)	Non-evolutionary failure search; failure monitors	A MAP-Elites map of where a latent world model is false-safe

Selection pressure on imagined safety ranks first

The ranking below weighs four things: novelty against the 2024–2026 check, feasibility before the 8 November freeze and before a late-January deadline on one RTX 5090 with about 24 CPU workers, fit with GECCO's tracks, and fit with the supervisor's brief and suggested papers.

Rank	Direction	Novelty	Feasible by 8 Nov	Feasible by late Jan	Best track	Role
1	Selection pressure $\times$ LeWM safety errors: evolve compact latent policies with penalty-free constraint handling, trace the exploitation curve, compare evolution-native mitigations at a matched real budget	High	Core curve plus 1–2 mitigations, ≥ 5 –10 seeds	Full mitigation comparison, a contrast with CEM-MPC, 10–20 seeds	NE (surrogate-assisted neuroevolution); EML as an alternative	Lead paper and thesis core
2	EC constraint handling vs penalties on real SafetyWalker2dVelocity-v1 , including a ROSARL-style return-range penalty and a δ -dial	Medium (old ranking rules; new benchmark and bridge)	Deb-rule ES, a penalty sweep and the PPO-Lagrangian reference	All arms: stochastic ranking, ϵ -constraint, AL-CMA-ES, ROSARL	EML	Ground-truth chapter; fallback paper
3	Safety-indexed MAP-Elites/CMA-MAE , with realised violation probability as the descriptor	High-medium	No	Reduced archive, corrected metrics	NE or CS	Stretch goal or second paper
4	A QD map of where LeWM is false-safe (predicted margin $\times$ gait or speed descriptors, re-evaluated)	Medium-high	As a diagnostic figure	Add-on to 1	NE or CS	Figure in paper 1
5	Constraint-handling ablation inside Safe CEM MPC	Medium	Already built	Folded into 1	NE	Baseline

Rank	Direction	Novelty	Feasible by 8 Nov	Feasible by late Jan	Best track	Role
6	LeWM-pre-screened illumination (SAIL/DA-QD-style, pessimistic cost acquisition)	High, with compound risk	No	Only if 1 and 3 both work	NE or CS	Post-thesis
7	ERL hybrid with a ROSARL-penalised gradient learner	Medium-high	No	Unlikely	EML	Future work

Direction 1 ranks first for three reasons. It is the only candidate that uses all of the student’s unusual assets at once: the LeWM, the probes as constraint oracle, exact snapshot branching as ground truth, and the measured **~45% false-safe rate at a 0.8 s horizon and zero margin** as a calibrated starting point. It is evolution-native, because population size, elite fraction, generations, re-evaluation and noisy-fitness selection are the experimental variables. And it fits the 2026 Neuroevolution scope, which explicitly lists “**Surrogate-assisted approaches**”, “**Multi-objective Neuroevolution**” and “**Applications to reinforcement... learning**” (NE track). Framing LeWM as a *dynamics surrogate* for constrained neuroevolution also places the work in an established GECCO lineage. Gaier et al.’s kernel-surrogate neuroevolution won a GECCO 2018 best paper (Gaier et al.), and Stork et al. followed at GECCO 2019 (Stork et al.). That literature always confirms promising candidates with real evaluations, and it names “false minima introduced by the surrogate” as the central risk (SAC-ERL). For the student, snapshot branching makes that confirmation almost free. The student’s own earlier finding, that boundary-focused predictor fine-tuning did not consistently help, also points this way: if fixing the model failed, fix the search.

Direction 2 ranks second because it is the cheapest and most literal answer to the supervisor’s brief, but its novelty is moderate. The ranking rules date to 2000, and the contribution would be the benchmark and the bridge, not a new method. Its specific risk is that ES needed **7.88× TRPO’s steps to reach TRPO’s final Walker2d score** (OpenAI-ES), and ARS reached the Walker2d threshold on **only 20% of seeds** (ARS). A real-environment ES therefore risks losing to PPO-Lagrangian on return at equal steps. That makes it a strong supporting chapter and a weak headline.

Direction 3 is the supervisor’s suggested line, and its novelty is real: no reward × cost QD archive for safe RL was found. Three problems keep it off the critical path before the freeze. First, the QD-RL norm is about **1 million evaluations per run**. The CMA-MEGA paper the supervisor suggested used 100 CPUs plus a GPU for **4–20 hours per run** (Tjanaka et al., GECCO 2022), and CPU-bound pyribs managed about 184 evaluations per second on 32 cores against roughly 30,000 for QDax on an A100 (QDax). Second, reviewers will raise the critique that one goal-conditioned policy can replace a bounded archive (Batra et al.), and CCPO already offers a threshold-conditioned safe policy. Third, MOME’s own Walker2d results show **no significant difference in global hypervolume between MOME and NSGA-II or SPEA2** ($p = 0.08\text{--}0.38$) (MOME). An archive therefore cannot claim a better frontier. It can claim guaranteed coverage at each safety level, plus diversity within each level.

On venue fit, the title screen of every 2023–2026 GECCO full paper and poster found **no evolutionary method on a CMDP safe-RL benchmark and no world model used as a neuroevolution fitness evaluator**. The closest items are Safe CMA-ES on benchmark functions, safe optimisation on binary spaces (2026), and a 2026 EML paper titled “Surrogate-Assisted Evolutionary Multi-Agent Reinforcement Learning with Adaptive Fitness Evaluation”, whose content was not readable (GECCO 2026 accepted papers). NE is small: it accepted 9, 9, 11 and 8 full papers in 2023–2026, and in 2025 it accepted **11 of 28 submissions (39%)** against EML’s 18 of 48 (SIGEVO chart). EML “strongly encourage[s]” comparisons with state-of-the-art non-evolutionary methods (EML track), which favours Direction 2 there. QD papers have historically appeared in both CS and NE. Analysis-led papers do succeed at GECCO: Zheng & Cheng’s EML 2023 study of population-assisted off-policy RL was an analysis paper (Zheng & Cheng), and Nisioti et al.’s benchmark paper was a 2025 best-paper nominee (Nisioti et al.). That supports a diagnosis-plus-remedy paper. Wits co-authors have recent NE (2024) and EML (2026 best-paper nominee) papers (2026 nominations), which

makes an internal pre-review with people who know those tracks possible. The submission risk is also low: under the 2026 rules there is **no maximum acceptance rate for poster papers submitted originally as full papers (2026 CFP)**.

The calendar decides the scope. Today is 3 October 2026. The freeze is 36 days away, and the thesis is due in late November. If 2027 repeats 2026’s pattern of a mandatory abstract on 19 January and a non-extensible full paper on 26 January, about eight working weeks remain for GECCO-specific work after the thesis. The plan below uses explicit gates so that a slipping harness triggers a pivot rather than a crisis.

Window	Dates	Work	Gate
Build	3–17 Oct	Imagination-ES harness (latent policy, branched fitness, probe oracle, paired snapshot re-evaluation); real-environment Deb-rule ES on state observations; throughput measurement; novelty sweeps	Gate 1 (17 Oct): paired imagined and real numbers exist for one configuration. If not, Direction 2 becomes the paper’s centre
Sweep	18 Oct–1 Nov	Exploitation curve over population size, generations, horizon, margin, noise and constraint rule; two mitigations (margin δ , top-k re-evaluation)	Gate 2 (1 Nov): a measurable trend with confidence intervals
Freeze	2–8 Nov	Core configurations to ≥ 10 seeds; PPO-Lagrangian reference evaluations; figures	Results freeze 8 Nov
Thesis	9–30 Nov	Writing	n/a
Extend	Dec–mid Jan	Full mitigation comparison at a matched budget; full constraint-handler crossing including ROSARL; contrast with CEM-MPC; 20 seeds where cheap; optional archive	Gate 3 (~20 Dec): archive goes into this paper or becomes a separate submission
Submit	~19–26 Jan 2027 (unconfirmed)	Abstract, full paper, anonymised code	n/a

One factorial design serves thesis and paper

The cleanest design crosses two factors. The first is the **evaluation substrate**: real MuJoCo, LeWM imagination, and LeWM imagination with a mitigation. The second is the **constraint-handling rule**. The real-MuJoCo row supplies ground truth and the PPO-Lagrangian comparison (Direction 2). The imagination rows supply the exploitation study (Direction 1). Reading across the grid answers the question no prior paper has posed: do the penalty-free rules the brief favours survive a learned model?

The lead experiment builds a safety τ table for a JEPA

The policy should follow Ha’s design: a linear map from the 192-dimensional LeWM embedding to Walker2d’s six actions (**PPGA**). That is about 1,200 parameters, close to Ha’s 867, which keeps full-covariance CMA-ES tractable. Larger MLP policies should switch to sep-CMA-ES, LM-MA-ES or OpenAI-ES, as JEDi did for Brax-sized genomes (**JEDi**). If a single embedding does not carry velocity, the input should be $[z_t, z_t - z_{t-1}]$, the Markovian latent that SLS² uses (**SLS²**). Initialising by regressing PPO-Lagrangian’s actions onto encoded latents starts evolution from a walker rather than from noise. It also sharpens the scientific question to this: *does further evolution in imagination improve the real policy, or only exploit the model?*

Fitness should follow MBPO. Score each candidate as the mean (or worst case) over K imagined 10-block, 0.8 s segments started from real latent states drawn from a bank of MuJoCo snapshots. This keeps every evaluation inside the validated horizon, because short rollouts branched from real data “scale[] to horizons that cause other model-based methods to fail entirely” (**MBPO**). The proposed design uses probe-decoded forward velocity as reward, and the probe-decoded health and speed rules (with margin δ) as feasibility predicates. A probe-decoded margin at the end of each segment captures consequences that arrive after it. The start-state bank should be refreshed from elites’ own real rollouts, which is the iterative loop Ha proposed but never ran on a hard task.

Ground truth comes from the student’s branching. Every candidate that selection keeps is executed closed-loop in MuJoCo from the same snapshot for the same 0.8 s. That yields **paired imagined and real outcomes on identical start states**, which is a stronger design than any of the cited exploitation studies had.

The pressure factors are population size λ , generations, elite fraction, imagined horizon (1–10 blocks, plus longer horizons to locate the breakdown), margin δ , and latent noise σ . The LeWM predictor is a deterministic regressor, so Ha’s temperature trick needs an analogue: Gaussian noise injected into predicted latents, scaled to the measured one-step residual.

The constraint-rule factor covers six options. Deb’s feasibility-first rule, credited as CCE, is the default (Deb 2000). Stochastic ranking, which “was designed to deal with the inherent shortcomings of a penalty function”, is the second (Runarsson & Yao). The ε -constrained method with a decreasing schedule is the third (Takahama & Sakai), and augmented-Lagrangian CMA-ES through pycma’s ConstrainedFitnessAL is the fourth (pycma). A fixed-penalty sweep and a ROSARL-style return-range penalty complete the set.

Three hypotheses organise the outputs.

H1, the optimiser’s curse for constraints. The false-safe rate among *selected elites* exceeds the 45% unconditional rate and rises with λ and generations. This is the constraint analogue of ASAR’s overgeneration result.

H2, feasibility-first is the most exploitable rule. Feasibility-first ranking concentrates selection on the optimistic tail of the constraint predictor. Stochastic ranking’s randomness partly shields against this. This is an inference, not a published result.

H3, amortised policies exploit more than replanning. An amortised evolved policy exploits more than per-step CEM-MPC with replanning, because replanning from true states limits how long an exploited error lasts, and ARC-Bench shows that replanning masks ranking defects (ARC-Bench).

The headline figure is the JEPA equivalent of Ha’s table: imagined against real violation and return, by σ and by pressure. A second figure tracks the Spearman correlation between imagined and real fitness across generations. This matters because open-loop rollout error ranked models worse than replay error did ($\rho = 0.774$ against 0.923), choosing a suboptimal estimator in 18 of 24 conditions (Raghavan & Singh).

Mitigations compete at a matched real-evaluation budget

The second half of the lead paper compares remedies under one accounting rule: every arm spends the same number of real MuJoCo steps, and the steps used to train LeWM are counted. The candidates map onto known model-based RL ideas but act through the evolutionary loop. The two cheapest are a **margin δ** calibrated from the student’s false-safe curve and **latent noise σ** , the analogue of Ha’s τ . The third is **disagreement pessimism**: a cheap ensemble (several probe heads, several predictor fine-tunes, or MC dropout) supplies a pessimistic cost estimate, combined with MOREL-style halting, which thresholds ensemble discrepancy into an absorbing HALT state (MOREL). Under feasibility-first ranking, halting converts “the model doesn’t know” into “infeasible” with no penalty weight. The fourth is **top-k real re-evaluation of elites through snapshots**. This is the transferability approach, which needed 11 physical trials against 200,000 simulated evaluations (Koos et al.), and it mirrors DA-QD’s selection of the 20 lowest-disagreement skills. The fifth adds **LeWM fine-tuning on the corrected data** to that re-evaluation.

The arms report real return, real violation probability, the imagined–real gap, and real steps consumed. This comparison is where a GECCO paper earns its “remedy” half. Its likely winner, real-verified elite selection, is also the remedy the surrogate-assisted neuroevolution literature already trusts.

The real-environment arm grounds the claims and carries the ROSARL bridge

Direction 2 runs ES and CMA-ES over linear or small MLP policies on the 17-dimensional state, through all six constraint rules. The comparison is OmniSafe PPO-Lagrangian, CPPPID, CPO and PPO. The reference numbers on SafetyWalker2dVelocity-v1 at 10^7 steps over 5 seeds are **PPO-Lagrangian 2982 ± 682 return**

and 13.5 ± 14.6 cost, and PPO 6240 ± 880 return and 903 ± 101 cost, against a cost limit of 25 ([OmniSafe benchmark](#)). The fixed-penalty sweep must be included so that the “hand-tuned penalty” baseline is not a strawman. Budgets should be reported at 10^7 steps (OmniSafe’s) and 10^8 steps (ES’s natural regime), with the fraction of seeds that succeed.

For the health-rule arm, the Walker2d healthy bonus should be removed, as SMBPO did when it treated MuJoCo termination as the unsafe event. Otherwise reward implicitly encodes safety ([SMBPO](#)). The official velocity cost (speed above 2.3415) should be reported alongside the health rule, because the student’s health rule is not the benchmark’s cost ([Safety-Gymnasium velocity](#)).

The bridge to the supervisor’s work is precise. Suppose each candidate is evaluated on a single episode and the episode terminates on violation. Then any penalty below every safe return ranks every violator below every safe candidate, which is exactly Deb’s first rule. ROSARL’s RLC 2026 version sets $R_{\text{unsafe}} = V_{\text{MIN}} - V_{\text{MAX}}$ and terminates on violation ([ROSARL](#), [RLJ 2026](#)), so under rank-based selection it differs from Deb only in two places. The first is how it orders violators: by pre-violation return, roughly time-to-violation, rather than by violation magnitude. The second is how it aggregates rollouts. With n rollouts and mean fitness it becomes a probability-weighted penalty, $(1-p) \cdot E[R \mid \text{safe}] + p \cdot R_{\text{unsafe}}$, which is no longer lexicographic in p .

An ablation of mean-fitness ROSARL-ES against lexicographic ($\hat{p}$, return) ranking and stochastic ranking therefore targets the brief’s “probability of safety violations” directly. With an indicator cost at first violation and termination, $E[\text{episodic cost}]$ equals $P(\text{violation})$, so “ $P(\text{violation}) \leq \delta$ ” needs no separate chance-constraint machinery. That links the work to chance-constrained EC ([Neumann & Witt](#)). It also gives the user an interpretable δ to set in place of λ . That matters because ROSARL’s own trade-off is steep: on PointGoal1 its cost fell to 0.08 against 0.62 for TRPO-Lagrangian, but its return fell to 1.87 against 10.17 ([RLJ 2026](#)).

The stretch archive must beat a conditioned policy, not just exist

If Gate 3 passes, the archive should be CMA-MAE in pyribs ([CMA-MAE](#); [pyribs](#)). The 1-D descriptor is realised violation probability or episodic cost, in 10–20 bins, with an optional second axis for speed. Return is the fitness, and the archive is seeded from PPO and PPO-Lagrangian checkpoints. The budget should be 10^4 – 10^5 evaluations rather than the 10^6 norm.

The results must use “corrected” archives: every elite is re-evaluated many times and placed back in its true cell, as PPGA does with **50 re-evaluations per agent** ([PPGA](#)). This matters more here than usual, because the descriptor is itself a noisy expectation and naive MAP-Elites fills cells with “lucky” elites ([Uncertain QD](#)).

The baselines are PPO-Lagrangian at k thresholds, NSGA-II compared on hypervolume, and a distilled cost-conditioned policy in the style of DCRL-ME or CCPO ([DCRL-ME](#)), which answers the conditioned-policy critique head-on. Constrained MAP-Elites feasibility and GuSS-style lexicographic replacement complete the set, and comparing cost as a descriptor with cost as a constraint is itself a clean ablation.

The hook to the CMA-MEGA paper the supervisor suggested is untested. CMA-MEGA’s measure gradient here is the gradient of expected cost, which a PPO-Lagrangian cost critic already estimates. A PPGA-style “safety-indexed CMA-MEGA” would therefore need no new estimator, but no paper has done this.

A cheaper diagnostic variant runs inside LeWM: archive by predicted margin $\times$ gait or speed descriptor, then re-evaluate each cell. This maps where false-safe errors cluster. It also tests whether they sit at gait-phase transitions, as the “kinematic, not dynamic” diagnosis of DreamerV3 walker-walk predicts ([Schäfer et al.](#)).

Statistics follow the GECCO QD-RL norm

Recent GECCO NE, EML and CS papers on RL use 5–20 seeds, typically 10. They report medians with IQR or 95% confidence intervals, and they use Mann-Whitney U or Wilcoxon rank-sum tests with Bonferroni or

Holm-Bonferroni correction. ASCII-ME used 20 seeds and compared methods at both equal runtime and equal evaluations (ASCII-ME); MEMES reported wall-clock time on identical hardware (MEMES). CRAX’s seed study is a warning about small seed counts. The 95% CI half-width relative to the mean shrank from about 90% at 2 seeds to 22% at 5, and small seed counts sometimes produced misleadingly narrow intervals (CRAX). The plan should report the interquartile mean with bootstrap CIs (reliable) and use Wilson or Clopper-Pearson intervals on every violation probability. Paired tests are appropriate on matched snapshots.

One arithmetic consequence deserves a sentence in the paper. With zero observed violations in n episodes, the 95% upper bound on $P(\text{violation})$ is about $3/n$ (the rule of three). Certifying a rate below 1% therefore needs roughly 300 clean episodes per final policy.

Budgets must appear on three axes: real steps (including LeWM’s training data), imagined steps, and wall-clock time. Otherwise a reviewer will argue that the evolutionary advantage disappears once all samples are counted. Safety Gym’s **cost rate over all of training** must count every real rollout of every population member (Safety Gym). Imagined evaluation is where a world-model method can win this metric, and naive real-environment ES is where it can lose.

Arm	Baselines	Primary metrics	Statistics
Lead (imagination)	LeWM Safe CEM-MPC; real-environment ES with the same rule; PPO-Lagrangian behaviour-cloned start; fixed-penalty CMA-ES	Elite false-safe rate; imagined–real Spearman ρ by generation; safety τ table; final real return, cost and $P(\text{violation})$; real steps, including LeWM data	≥ 10 seeds before the freeze, 20 for GECCO; paired tests on snapshots; Holm correction
Real environment	OmniSafe PPO-Lagrangian, CP-POPID, CPO, PPO; fixed-penalty sweep	Return and cost against limit 25; fraction of seeds feasible; cost rate over every rollout; $P(\text{violation})$ with Wilson CI	Mann-Whitney U with Holm; IQM with bootstrap CI
Archive (stretch)	PPO-Lagrangian at k thresholds; NSGA-II; distilled conditioned policy; Constrained MAP-Elites; GuSS rule	Corrected QD-score and coverage; dial curve; 2-D hypervolume; descriptor fidelity per bin; monotonicity; real steps to produce the archive	As above

On compute, extrapolating from the student’s 24 workers at 8,000 steps/s each, and assuming 50–75% efficiency after policy inference and stragglers, gives about 100–150k effective steps/s. That puts 10^8 steps at 11–17 minutes and a grid of 4 methods $\times$ 10 seeds $\times$ 10^8 steps at about **7.5–11 hours**, while a QD run of 10^9 steps takes 2–3 hours. These are estimates, not measurements, but even a several-fold shortfall leaves the plan intact. Compute is not the bottleneck; calendar time is.

The lowest-risk stack keeps the vendored Safety-Gymnasium simulator on CPU, uses PyTorch-native optimisers (pycma, EvoTorch, pyribs) and runs LeWM on the 5090. Porting to CRAX’s GPU Walker2D Velocity task would change the simulator, the thresholds and the baselines, and CRAX is an unreviewed preprint with a 15-star codebase (CRAX repo).

Exploitation is the finding, and also the threat

The central risk is close to certain: evolution will exploit LeWM. Ha’s agent learned to move so that the dream’s monsters “never shoots a single fireball” (Ha & Schmidhuber). Evolution’s history is full of fitness loopholes (Lehman et al.), and a 2026 backdoor study shows CEM re-discovering whatever the model’s local dynamics reward (Backdoors). The design answer is to make exploitation the measured quantity rather than a hidden confound. That only works if at least one remedy beats the baseline. Otherwise the paper is a negative result, which is harder to place.

There is a second-order risk that most proposals miss: **the mitigations themselves fail in the region evolution explores**. Ensemble disagreement ranks out-of-distribution errors poorly (Intervention gap), and

UNISafe’s authors concede that their ensemble “may not faithfully capture” true uncertainty (UNISafe). Conformal margins reduce false-safes at the cost of conservativeness (Kim et al.). This is why real re-evaluation through snapshots must be the backstop rather than one arm among equals.

Long-horizon error is the second structural risk. LeWM’s authors state that “auto-regressive rollouts accumulate prediction errors as the horizon grows” and restrict planning to short horizons (LeWM). Hierarchy on a frozen LeWM helps only “when high-level search remains compatible with the low-level controller” (Caselli et al.). Most directly relevant to a Walker health rule, a DreamerV3 walker-walk model’s imagined kinematic-consistency error **stayed flat while real reward collapsed** across a friction sweep, and failures appeared beyond the gait period (Schäfer et al.).

Coverage compounds both risks. If LeWM’s data came mostly from competent PPO and PPO-Lagrangian rollouts, falls are under-represented, and evolution will push into exactly those states. Seeding the snapshot bank with perturbed and failing rollouts is cheap insurance. Theory supports keeping policies smooth: return-error bounds for training in imagination tighten as the Lipschitz constants of the dynamics, reward and policy shrink (Timor et al.). That favours linear policies with bounded weights.

Risk	Evidence	Mitigation	Residual
Elites exploit false-safe predictions	Ha’s τ table; ASAR pool-size collapse	Treat as the measured quantity; top-k snapshot re-evaluation; selection on lower confidence bounds	A paper of negative results only if no remedy wins
Disagreement misses out-of-distribution error	Intervention gap	Real re-evaluation as backstop; MOREL-style halting as one arm only	Pessimism may become over-conservative
Long-horizon drift; imagination is kinematic, not dynamic	LeWM; Schäfer et al.	Branched segments of ≤ 0.8 s; terminal probe margin; model selection by branched error	Delayed consequences are under-scored
ES sample inefficiency and seed fragility	$7.88 \times$ TRPO; 20% seed success on Walker2d (ARS)	Behaviour-cloned initialisation; report success fraction; ≥ 10 seeds	May lose to PPO-Lagrangian on return
Feasibility rules converge prematurely; noisy costs	Mezura-Montes & Coello; Uncertain QD	Stochastic ranking and ϵ arms; feasibility from confidence intervals; resampling	No published effect sizes for RL
Training-time violations from population rollouts	Safety Gym cost rate	Count every rollout; imagination incurs no real cost	Real-environment ES looks worst here
Reviewer novelty objections	CCE, RCE, CCEM; CCPO; Batra et al.	Present Safe CEM as a baseline; include a distilled conditioned policy	n/a
Benchmark outside the safe world-model norm	SafeDreamer uses Point/Car Goal; no world-model paper reports Walker velocity	Report the official velocity cost; cite SMBPO for health as the constraint	One environment; add Hopper velocity if time allows
Tooling	Safety-Gymnasium pins MuJoCo 2.3.3 and lacks Python 3.11 (repo); RTX 5090 needs CUDA ≥ 12.8 for PyTorch (forum)	Stay on the existing CPU stack; no JAX port before the freeze	n/a
Being scooped in a fast-moving JEPA field	SLS ² (Jun), ASAR (Jul), FARM and ARC-Bench (Sep 2026)	Post to arXiv early (GECCO allows it); weekly arXiv alerts	n/a

Seven checks gate the commitment

Several claims in this report rest on negative search results or unverified metadata. Each of the following should be closed before Gate 1 on 17 October, and the novelty checks repeated before submission.

Check	Why it matters	State on 3 Oct 2026
Read the 2026 GECCO papers known only by title in the ACM DL: Tsinghua’s surrogate-assisted EMARL; Ajani et al.’s dimensionality-reduced neuroevolution of deep RL; posters “QDA-VC”, “MO-CEM-RL” and “A Multi-Factorial PSO for Balancing Safety and Efficiency”	These are the closest venue-level competitors	Title only; the ACM DL and dblp were blocked (2026 posters)
Follow Semantic Scholar citation trails for Ha & Schmidhuber, GuSS, DA-QD, CCPO, Fioravanzo & Iacca, LeWM, SLS ² and SafeDreamer, and run an arXiv sweep for September–October 2026	The “no paper found” claims are negative results	Not done; the arXiv API rate-limited one session
Resolve possibly overlapping preprints: JEPA-TTT (2610.00722); the claim that Spectral-Target JEPA uses CEM with a collision-penalty term (2609.04264); ARC-Bench’s date mismatch; the details of PIGDreamer and FOSP	Possible prior “penalty CEM on a JEPA”	Seen only in snippets
Confirm the ROSARL citation with the supervisor	The brief says RLC 2024. arXiv 2306.00035 lists no venue (arXiv), the RLJ 2024 issue has no Nangue Tasse paper (RLJ 2024), and the method appears as “An Unreasonably Simple Approach to Safe RL” at RLC 2026 (RLJ 2026)	Evidence points to RLC 2026
Confirm GECCO 2027 dates, tracks and workshops, and whether ACM Open covers Wits for the APC	The deadline plan assumes the 2026 pattern	Only the venue is confirmed (Kraków, 12–16 July 2027, in person only); the dates and call pages still show 2026 content (GECCO 2027)
Measure local throughput: CPU steps/s including policy inference; rendering plus encoding for latent-policy evaluation; batched LeWM imagined steps on the 5090	Every budget above is extrapolated	The figure of 8,000 steps/s per process is unverified with inference overhead
Re-verify numbers before quoting them: SLS ² ’s results come from a tool summary; PPGA’s Walker2d table; Runarsson & Yao’s Pf of 0.45; conflicting venues for EPLS, DA-QD, LAMBDA and S-3PO	Errors in a related-work section draw reviewer scrutiny	Flagged during research, not yet re-checked

Two of these deserve emphasis. The ROSARL discrepancy is more than bibliographic. The RLC 2026 version, with its running $V_{\text{MIN}} - V_{\text{MAX}}$ penalty and its “maximally safe” objective, is the one whose mechanics the Deb-rule bridge uses, so the paper should cite the version the supervisor considers canonical. The fallback venues also depend on 2027 renewals that are unknown. The 2026 Surrogate-Assisted Evolutionary Optimisation workshop explicitly listed “Surrogate-assisted identification of the feasible region” (workshop), and EC for Autonomous Cyber-Physical Systems listed “Safety assurance” and “Reducing the simulation-reality gap” (workshop). Both had deadlines in late March or early April, well after the thesis.

Conclusion

The research landscape turns the obvious proposal upside down. Building a safe planner on a JEPA world model is no longer open ground: SLS², SafeDreamer and the latent-filter papers occupy it, and the student’s Safe CEM restates a rule published in 2018. What remains open is a question only an evolutionary-computation researcher with exact ground truth can answer cheaply. How much of a constraint predictor’s error does selection pressure harvest, and which changes to the *selection operator*, rather than to the model, give it back? The student’s prior results (a 45% base false-safe rate, a 75% dynamics attribution on Push-T, and fine-tuning that did not fix it) stop being limitations and become the calibration curve for that study.

The tension at the core of the supervisor’s brief is also the paper’s most interesting hypothesis. Penalty-free

constraint handling (Deb rules, and ROSARL's minmax penalty under rank selection, which collapses into the same rule) is attractive because it is scale-free and needs no λ . But a rule that ranks every predicted-feasible candidate above every predicted-infeasible one is exactly the rule that rewards optimistic errors in the feasibility predictor. If H2 holds, the safest-looking choice for optimisation becomes the riskiest choice under a learned model, and the remedy has to come from evolution's own toolkit: re-evaluation, noise-aware ranking and pessimistic selection. That is a GECCO-native result whichever way the data falls.